

A literature-implied no- c_0 classification for Le Merdy's $C(\mathbb{T})$ -extension question

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Source Question

In γ -bounded representations of amenable groups [1, p. 27], Le Merdy asks:

When X does not have property (α) , find a characterization of bounded invertible operators $T : X \rightarrow X$ such that $\pi : k \in \mathbb{Z} \mapsto T^k$ extends to a bounded homomorphism $C(\mathbb{T}) \rightarrow B(X)$.

would be κ -bounded as well, by Theorem 4.4 and the cotype assumption. $\square$

According to the above discussion, Proposition 6.6 applies on S^p for any $1 \leq p \neq 2 < \infty$, as well as on any space of the form $\text{Rad}(X)$ when X does not have property (α) but has a finite cotype. This leads to the following general question:

When X does not have property (α) , find a characterization of bounded invertible operators $T : X \rightarrow X$ such that $\pi : k \in \mathbb{Z} \mapsto T^k$ extends to a bounded homomorphism $C(\mathbb{T}) \rightarrow B(X)$.

Supporting Literature

Kriegler and Le Merdy [2] prove general facts about bounded $C(K)$ -representations. Lemma 3.8 says that if $u : C(K) \rightarrow B(X)$ is a representation and $T = u(\kappa)$, where $\kappa(z) = z$, then $\sigma(T) \subset K$, the representation vanishes on the ideal of functions zero on $\sigma(T)$, and therefore factors through a representation $v : C(\sigma(T)) \rightarrow B(X)$ with spectral mapping.

Their Remark 3.10 and Corollary 3.11 identify the weak-compactness condition that turns such a calculus into scalar-type spectrality. In particular, when X contains no copy of c_0 , bounded maps from a $C(K)$ -space into X are weakly compact, and the scalar-type obstruction disappears.

and Lemma 3.8, taking the implication (2) $\Rightarrow$ (1), also follows from Lemma 3.8. $\square$

Remark 3.10. Let A be a sectorial operator with a uniformly bounded H^∞ -calculus, and let $T = (1 + A)^{-1}$. It follows from Lemma 3.8 and the proof of Proposition 3.9 that there exists a representation

$$v: C(\sigma(T)) \longrightarrow B(X)$$

satisfying $v(\kappa) = T$ (where $\kappa(z) = z$), such that $\sigma(v(f)) = f(\sigma(T))$ for any $f \in C(\sigma(T))$ and v is an isomorphism onto its range. Also, it follows from the proof of Lemma 3.8 that T^* is a scalar-type operator of class X , in the sense of [12, Def. 5.14].

Next according to [12, Thm. 6.24], the operator T (and hence A) is a scalar-type spectral operator if and only if for any $x \in X$, the mapping $C(\sigma(T)) \rightarrow X$ taking f to $v(f)x$ for any $f \in C(\sigma(T))$ is weakly compact.

Corollary 3.11. *Let A be a sectorial operator on X , with $\omega(A) = 0$, and assume that X does not contain a copy of c_0 . Then A admits a uniformly bounded H^∞ -calculus if and only if it is a scalar-type spectral operator.*

Proof. The ‘only if’ part follows from the previous remark. Indeed if X does not contain a copy of c_0 , then any bounded map $C(K) \rightarrow X$ is weakly compact [9, VI, Thm. 15]. (See also [29] and [8] for related approaches.) The ‘if’ part follows from [15, Prop. 2.7] and its proof. $\square$

Implied Classification

Theorem. *Let X be a Banach space containing no copy of c_0 , and let $T \in B(X)$ be invertible. Then the following are equivalent.*

1. *The representation $k \in \mathbb{Z} \mapsto T^k$ extends to a bounded unital homomorphism $u: C(\mathbb{T}) \rightarrow B(X)$ satisfying $u(\kappa) = T$.*
2. *$\sigma(T) \subset \mathbb{T}$ and T is a scalar-type spectral operator.*

Proof. Assume (1). By Lemma 3.8 of [2], u factors through a bounded representation $v: C(\sigma(T)) \rightarrow B(X)$ with $v(\kappa) = T$. For each $x \in X$, the orbit map

$$C(\sigma(T)) \longrightarrow X, \quad f \longmapsto v(f)x$$

is bounded. Since X contains no copy of c_0 , every bounded operator from a $C(K)$ -space into X is weakly compact. By the scalar-type criterion recorded in Remark 3.10 of [2], T is a scalar-type spectral operator. Also $\sigma(T) \subset \mathbb{T}$ because $T = u(\kappa)$ and κ has range $\mathbb{T}$.

Conversely, if T is scalar-type spectral and $\sigma(T) \subset \mathbb{T}$, the scalar-type spectral resolution gives the standard bounded continuous functional calculus $v: C(\sigma(T)) \rightarrow B(X)$ with $v(\kappa) = T$. Composing restriction $C(\mathbb{T}) \rightarrow C(\sigma(T))$ with v gives the desired bounded homomorphism $C(\mathbb{T}) \rightarrow B(X)$. $\square$

Scope

This is a partial classification of the source question: it covers the no- c_0 subcase, including many finite-cotype non-property- (α) examples such as Schatten S^p , $1 \leq p < \infty$, $p \neq 2$. It does not settle arbitrary spaces without property (α) , especially spaces containing c_0 , and it does not classify when the resulting homomorphism is R - or γ -bounded.

References

- [1] C. Le Merdy, *γ -bounded representations of amenable groups*, arXiv:1003.1587.
- [2] C. Kriegler and C. Le Merdy, *Tensor extension properties of $C(K)$ -representations and applications to unconditionality*, arXiv:0901.1025.